

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1707

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I _____ V.Lasya(192524186) _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: v.lasya

Descriptive Written Test

- 1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.**

Smart Home Automation System Using Intelligent Agents

A smart home automation system uses Artificial Intelligence (AI) to automatically control devices such as lights, air conditioners, heaters, and security systems. It continuously receives information from sensors like motion detectors, temperature sensors, cameras, and clocks, and takes appropriate actions based on user behavior and environmental conditions. Intelligent agents help the system make decisions efficiently.

1. Simple Reflex Agent

A Simple Reflex Agent makes decisions based only on the current input without considering previous events or future consequences. It follows predefined condition-action rules.

Working in the Smart Home:

- If motion is detected in a room, turn on the lights.
- If no motion is detected for 10 minutes, turn off the lights.
- If the temperature exceeds 30°C, switch on the air conditioner.
- If a door opens unexpectedly at night, trigger the alarm.

Advantages:

- Fast response.
- Easy to implement.
- Requires very little computation.

Limitations:

- Cannot remember previous states.
 - Cannot adapt to user habits.
 - Performs poorly in complex or changing environments.
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2. Model-Based Reflex Agent

A Model-Based Reflex Agent maintains an internal model of the environment. It remembers previous states and uses both current sensor data and stored information to make better decisions.

Working in the Smart Home:

- Remembers whether someone is already inside a room before switching lights off.
- Tracks temperature changes over time before adjusting heating or cooling.
- Identifies whether doors were previously locked before raising a security alert.
- Uses room occupancy history to avoid unnecessary lighting changes.

Advantages:

- Handles partially observable environments.
- Makes more accurate decisions.
- More reliable than simple reflex agents.

Limitations:

- Requires additional memory.
 - More computationally expensive.
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3. Goal-Based Agent

A Goal-Based Agent chooses actions that help achieve a predefined objective. Instead of simply reacting, it evaluates different actions and selects the one that best satisfies the goal.

Working in the Smart Home:

- Maintaining comfortable room temperature.
- Reducing electricity consumption.
- Ensuring maximum home security.

Examples:

- Decides whether opening windows is better than turning on the air conditioner.
- Delays appliance operation until off-peak electricity hours.
- Chooses the most energy-efficient method to maintain desired room temperature.

Advantages:

- Makes intelligent decisions.
- Considers multiple possible actions.
- Flexible in dynamic environments.

Limitations:

- Requires planning algorithms.
 - Slower than reflex agents due to decision-making.
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4. Utility-Based Agent

A Utility-Based Agent not only achieves goals but also selects the action that provides the highest overall benefit based on a utility function.

The utility function may consider:

- User comfort
- Energy efficiency
- Security
- Cost of electricity
- User preferences

Working in the Smart Home:

- If electricity prices are high, slightly increase AC temperature to save energy while maintaining comfort.
- Adjust lighting brightness based on available natural sunlight.
- Balance security and convenience by recognizing trusted family members through facial recognition.
- Learn preferred sleeping temperature and automatically maintain it every night.

Advantages:

- Optimizes multiple objectives simultaneously.
- Learns user preferences.
- Provides the best overall experience.

Limitations:

- Complex to design.
- Requires more computational power and learning algorithms.

Best Approach for the Smart Home System

A hybrid approach combining Model-Based, Goal-Based, and Utility-Based Agents is the most effective solution.

- **Model-Based Agent** maintains the current state of the home by remembering occupancy, temperature history, and device status.
- **Goal-Based Agent** ensures objectives such as comfort, energy efficiency, and security are achieved.
- **Utility-Based Agent** selects the optimal action by balancing user comfort, electricity consumption, security, and user preferences.

The **Simple Reflex Agent** can still be used for immediate responses, such as instantly turning on lights when motion is detected or activating alarms during emergencies.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice

Robot Navigation in an Unknown Maze Using Uninformed Search Strategies

A robot placed in an **unknown maze** must explore its surroundings and find a path to the exit without having prior knowledge of the maze layout. Since the robot does not know where the exit is or which path is best, it uses **uninformed (blind) search algorithms**. These algorithms search the state space without any heuristic information and rely only on the problem definition.

1. Uniform Cost Search (UCS)

Uniform Cost Search (UCS) expands the node with the **lowest cumulative path cost** first. It guarantees finding the least-cost path if all step costs are positive.

Working in the Maze

- The robot starts from the entrance.
- It records the cost of reaching each location.
- At every step, it chooses the unexplored path with the smallest total cost.
- The process continues until the robot reaches the exit.

Example:

If different paths have different movement costs (stairs, rough terrain, slippery floor), UCS always selects the cheapest path.

Advantages

- Finds the optimal (least-cost) solution.
- Complete, provided all path costs are positive.
- Suitable for weighted mazes.

Disadvantages

- Requires large memory to store all explored nodes.
- Slow when many nodes have similar costs.
- Computationally expensive for large mazes.

Complexity

- **Time Complexity:** $O(b^{1+\lceil C^*/\epsilon \rceil})$
- **Space Complexity:** $O(b^{1+\lceil C^*/\epsilon \rceil})$

Where:

- **b** = branching factor
 - **C*** = cost of the optimal solution
 - **ε** = minimum step cost
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2. Iterative Deepening Search (IDS)

Iterative Deepening Search (IDS) repeatedly performs Depth-First Search with increasing depth limits until the goal is found.

Working in the Maze

The robot searches:

- Depth 0
 - Depth 1
 - Depth 2
 - Depth 3
- and continues increasing the depth until it discovers the exit.

Advantages

- Uses very little memory.
- Complete for finite search spaces.
- Finds the shallowest solution when all step costs are equal.
- Combines the advantages of BFS and DFS.

Disadvantages

- Repeats exploration of upper-level nodes.
- Not suitable when path costs differ.
- Can be slower because of repeated searches.

Complexity

- Time Complexity: $O(b^d)$
- Space Complexity: $O(bd)$

Where:

- **b** = branching factor
- **d** = depth of the shallowest solution

Intelligent Agents in the Maze Problem

Different intelligent agents behave differently while navigating the maze.

1. Simple Reflex Agent

- Makes decisions using only the current sensor input.
- Example: Move forward if the path is clear; otherwise, turn.

Limitation: Cannot remember previously visited locations, so it may repeatedly enter loops.

2. Model-Based Agent

- Maintains an internal map of explored areas.
- Remembers visited paths and blocked routes.
- Avoids revisiting the same locations.

Advantage: Much more efficient in unknown environments.

3. Goal-Based Agent

- The goal is to reach the maze exit.
- Evaluates different possible paths and selects actions that move closer to the goal.

Advantage: Makes decisions based on achieving the objective rather than simply reacting.

4. Utility-Based Agent

- Chooses the path that maximizes overall utility.
- Considers factors such as:
 - Travel distance
 - Energy consumption
 - Time taken
 - Safety

Advantage: Produces the most efficient overall solution.

Effect of Agent Design on Decision-Making

The design of an intelligent agent directly affects how the robot navigates the maze.

- **Simple Reflex Agent:** Fast decisions but poor performance in unknown environments because it has no memory.
- **Model-Based Agent:** Uses memory to avoid loops and improve navigation.
- **Goal-Based Agent:** Plans actions that lead toward the exit.
- **Utility-Based Agent:** Chooses the best path by balancing distance, energy, and time.

As the agent becomes more intelligent, its decision-making becomes more effective and efficient.

Best Combination of Agent and Search Strategy

The most effective combination is a **Model-Based Goal-Based Agent with Uniform Cost Search (UCS)**.

Justification

- The **Model-Based Agent** stores information about explored areas and prevents revisiting the same paths.

- The **Goal-Based Agent** continuously plans actions that lead toward the maze exit.
- **Uniform Cost Search** guarantees the least-cost path, making it ideal when movement costs vary.

If the maze has **equal movement costs** and **memory is limited**, a **Model-Based Agent with Iterative Deepening Search (IDS)** is a better choice because IDS requires much less memory while still guaranteeing a solution.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand